# Mth633 (Quiz 4)

# Most Important Mcqs Final term



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Time Left 69 sec(s)

MTH633 - Group Theory (Quiz 4)

Quiz Start Time: 01:06 PM, 07 July 2025

**Question # 5 of 5 ( Start time: 01:08:38 PM, 07 July 2025 )** **Total Marks:** 1

The following are three equivalent conditions for a subgroup H of a group G to be a normal *subgroup* of G:

**1.** $ghg^{-1} \in H$ for all $g \in G$ and $h \in H$.

**2.** $gHg^{-1} = H$ for all $g \in G$.

**3.** $gH = Hg$ for all $g \in G$.

**Select the correct option**

Reload Math Equations

- (●) True
- ( ) False

Click to Save Answer & Move to Next Question

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MTH633 - Group Theory (Quiz 4)

Quiz Start Time: 01:06 PM, 07 July 2025

**Question # 4 of 5 ( Start time: 01:08:10 PM, 07 July 2025 )** **Total Marks:** 1

Let G be a group and H be a subgroup of G then H is normal iff $(aH)(bH) \neq (ab)H$, for all $a, b \in G$.

**Select the correct option**

Reload Math Equations

- ( ) True
- (●) False

Click to Save Answer & Move to Next Question

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Quiz Start Time: 01:06 PM, 07 July 2025

Question # 3 of 5 ( Start time: 01:07:45 PM, 07 July 2025 )

Total Marks: 1

The trivial subgroup $N = \{0\}$ of $Z$ is not a normal subgroup of $Z$.

Select the correct option

Reload Math Equations

- True
- False

Click to Save Answer & Move to Next Question

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MTH633 - Group Theory (Quiz 4)

Quiz Start Time: 01:06 PM, 07 July 2025

Question # 2 of 5 ( Start time: 01:07:10 PM, 07 July 2025 )

Total Marks: 1

Let G be a group and H be a subgroup of G then H is normal iff $(aH)(bH) = (ab)H$, for all $a, b \in G$.

Select the correct option

Reload Math Equations

- False
- True

Click to Save Answer & Move to Next Question

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Time Left 74 sec(s)

MTH633 - Group Theory (Quiz 4)

Quiz Start Time: 01:06 PM, 07 July 2025

Question # 1 of 5 ( Start time: 01:06:37 PM, 07 July 2025 )

Total Marks: 1

Let a projection map $f : Z_4 \times Z_2 \to Z_4$ given by $f(x, y) = x$ is a homomorphism where $\ker(f)$ is $\{0\} \times Z_2$, then which of the following, the factor group $Z_4 \times Z_2/\{0\} \times Z_2$ is isomorphic to?

Select the correct option

Reload Math Equations

- $Z_2$
- $\{0\}$
- $Z_4 \times Z_2$
- $Z_4$

Click to Save Answer & Move to Next Question


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